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# AI for Bharat Hackathon

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## Problem Statement :

Startups often lack SRE expertise, causing infrastructure misconfigurations and rising cloud costs to go undetected until user impact.
There is no intelligent system that continuously interprets DevOps telemetry to proactively recommend cost-performance optimizations.

## AutoPilotAI: Your AI SRE Team in a Box

**The Problem:** Indian startups move fast shipping features, scaling users, burning through AWS credits. But when infrastructure costs explode or performance degrades, there's no SRE team to catch it. By the time developers notice, users are already complaining and money is already wasted.

**Our Solution:** AutoPilotAI is a multi-agent AI system that acts as your dedicated SRE team. It doesn't just monitor, it understands. Instead of showing you "CPU: 78%", it tells you:
*"Your Celery worker pool is mis-sized for your Redis throughput. You're experiencing job starvation. Increase workers from 8 to 12. This will cost an extra ₹2,400/month but save ₹18,300/month by preventing Redis over-provisioning."*

**How It Works:** Six specialized AI agents work together, each an expert in one domain (observability, infrastructure, databases, cost, CI/CD). A Planner Agent orchestrates them, using Claude Sonnet for deep analysis and Amazon Q to generate custom tools on-the-fly.

## How AutoPilotAI is Different from Existing Solutions ?

### Traditional Monitoring vs AutoPilotAI:

### 1. Raw Metrics → Semantic Intelligence

**Them:** "CPU: 78%, Memory: 6.2GB, Redis connections: 847"
**Us:** "Your worker pool can't keep up with Redis queue depth. Job processing is delayed by 12 seconds on average. Fix: Scale workers to 12. Impact: ₹18,300/month savings.“

### 2. Reactive Alerts → Predictive Prevention

**Them:** Alert you AFTER the database crashes
**Us:** "Your query load is growing 15% weekly. At this rate, you'll hit connection limits in 8 days. Here's the fix before it breaks.“

**3. Single-Purpose Tools → Multi-Agent Intelligence**

**Them:** Separate tools for monitoring, cost, databases—you connect the dots
**Us:** 6 AI agents collaborate automatically. When one spots an issue, others analyze cost impact, root cause, and deployment risk together.

**4. Generic Recommendations → Cost-Aware Actions**

**Them:** "Consider upgrading your instance"
**Us:** "Upgrade to r6g.xlarge: +₹8,200/month, but saves ₹24,000/month in Redis costs. ROI: 192%. Migration script attached.“

**5. Static Monitoring → Self-Evolving System**

**Them:** You configure what to monitor
**Us:** AI generates new monitoring tools using Amazon Q when it detects gaps. System extends itself based on your infrastructure needs.

## 6. Global Solutions → India-First Design

**Them:** Costs in USD, optimized for US/EU workloads

**Us:** Every number in ₹. Optimized for AWS India pricing. Built for startup budgets, not enterprise wallets.

The Bottom Line: Existing tools show you problems. AutoPilotAI prevents them, explains them in business terms, and tells you exactly what to do—with the cost-benefit analysis already done.

# How will it be able to solve the problem?
## The Six AI Agents (Orchestrated by Planner Agent)

**1. Observability Agent - Your Infrastructure Translator**

Converts metrics into business insights with context
Detects anomalies before they become incidents
Attributes performance issues to specific components
Example: "Redis fragmentation at 2.8x causing 340ms latency spike → Restart recommended“

**2. Infrastructure Agent - Configuration Guardian**

Analyzes Dockerfiles and ECS configs for optimization
Detects infrastructure drift automatically
Right-sizes worker pools based on actual throughput
Example: "Your Celery workers are starving—scale from 8 to 12 for optimal throughput“

**3. Database Agent - Query Optimizer**

Analyzes PostgreSQL EXPLAIN plans

Recommends missing indexes with DDL statements

Optimizes Redis eviction policies

Example: "Add index on users(email, created_at) → 87% faster queries, 0.2s → 0.026s“

**4. Cost Agent - Your CFO's Best Friend**

Calculates cost impact of every change in ₹

Identifies over-provisioned resources

Ranks optimizations by ROI

Example: "Right-size RDS from db.r5.2xlarge → db.r5.xlarge: Save ₹31,200/month, 0% performance impact"

**5. CI/CD Agent - Build Performance Guardian**

Tracks build time trends across commits

Detects regressions and attributes to specific changes

Predicts failures before they happen

Example: "Build time increased 40% after commit abc123 (added heavy dependencies)“

**6. Planner Agent - The Orchestrator**

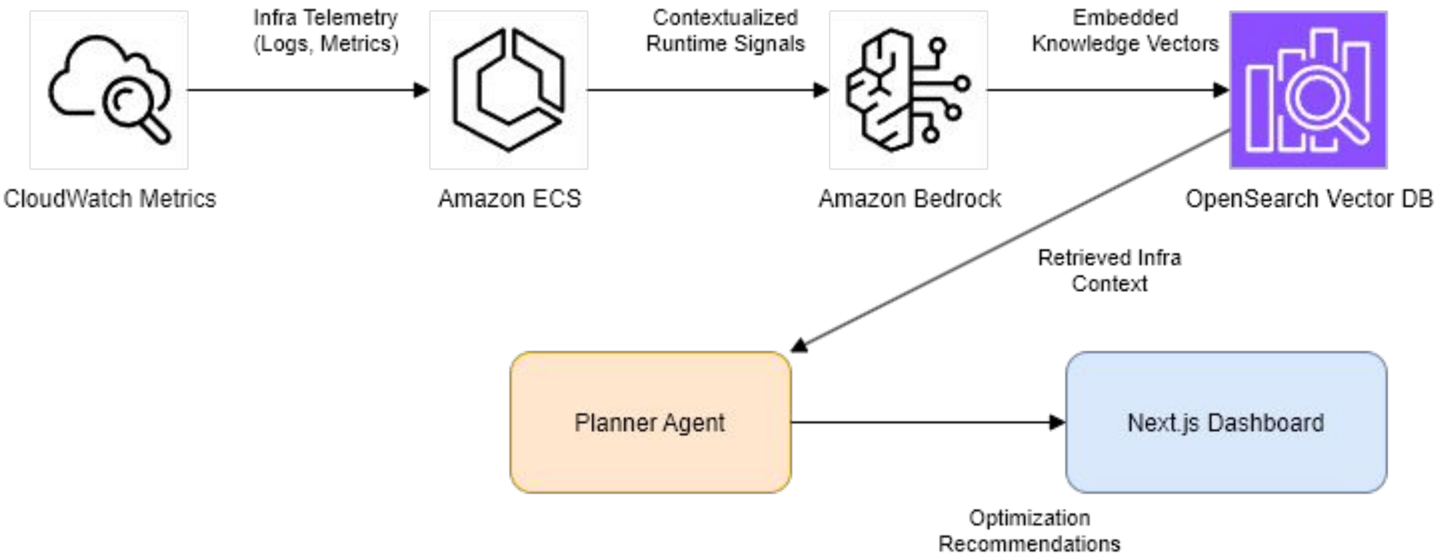
Coordinates all specialized agents

Synthesizes multi-agent insights

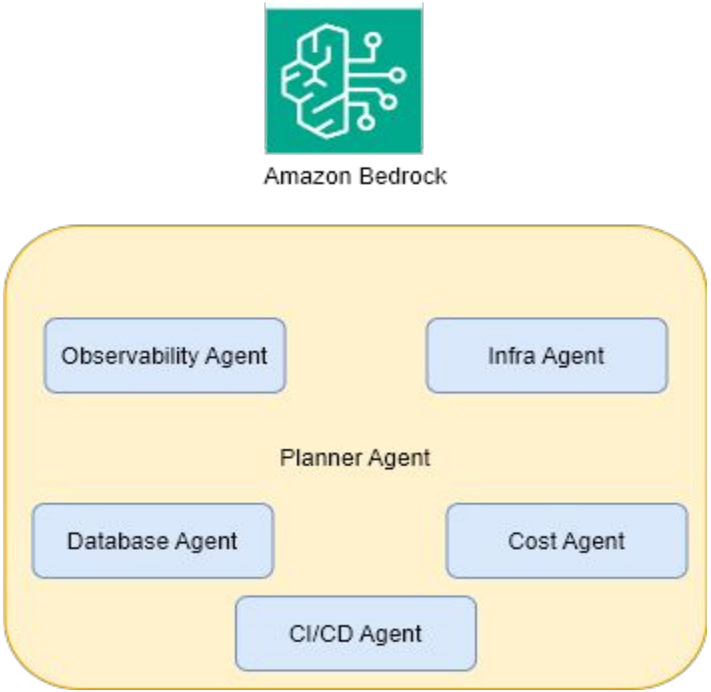
Prioritizes actions by business impact

Uses Amazon Q to generate custom tools on-demand

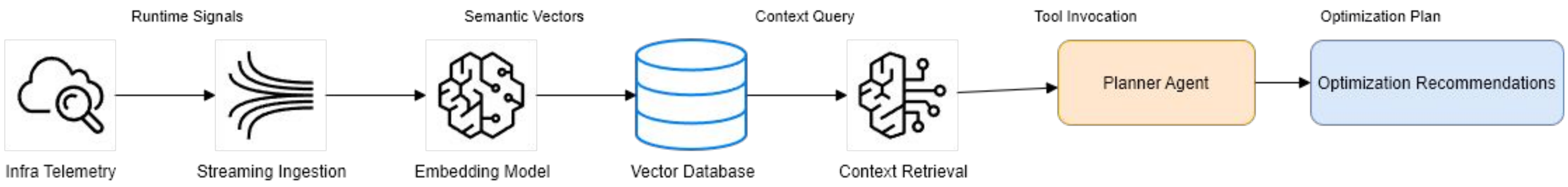
# Architecture of AutoPilot-AI



# Agent Layer of AutoPilot-AI



# RAG Pipeline of AutoPilot-AI



# Technologies to be used in the solution:

- **AWS AI/ML Core**
  - Amazon Bedrock - Multi-agent orchestration
  - Claude Sonnet 4.6 - Root-cause analysis & insights
  - Bedrock Knowledge Bases - RAG system
  - Titan Embeddings - Configuration indexing
  - Amazon Q Developer - Dynamic tool generation
- **AWS Infrastructure**
  - CloudWatch - Metrics & logs
  - ECS - Container data
  - RDS/ElastiCache - Database metrics
  - Billing API - Cost analysis (INR)
  - S3 - Config storage
  - IAM - Security & access
- **External Integrations**
  - GitHub API - Commit & CI/CD analysis
  - PostgreSQL - Query plan analysis
  - Redis - Performance metrics
- **Development**
  - Python 3.11+ - Implementation
  - Hypothesis - Property-based testing
  - Docker - Containerization
  - Terraform - Infrastructure as Code

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## Thank You